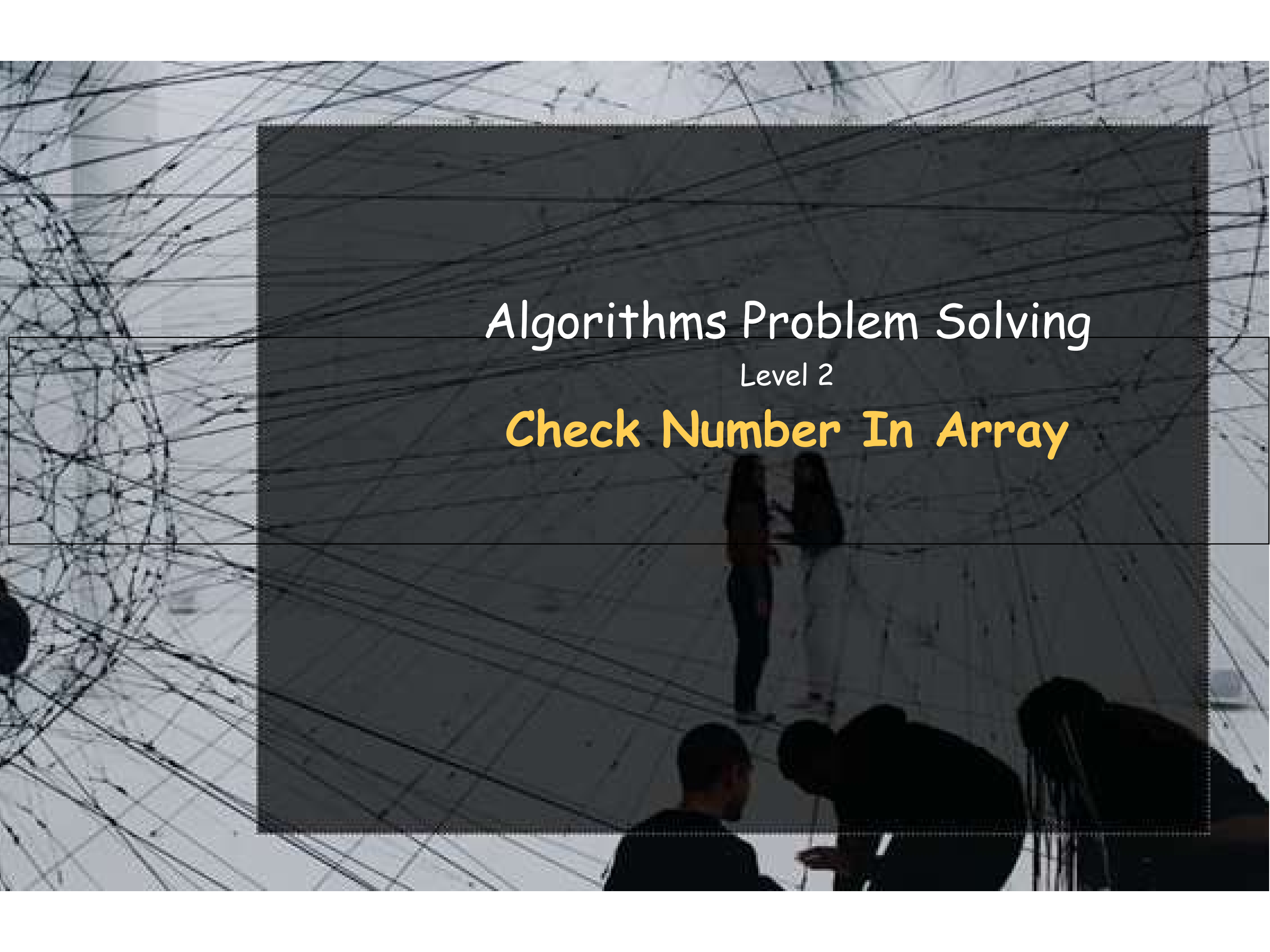


Algorithms Problem Solving Level 2

Check Number In Array

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the lower portion of the image, the silhouettes of several people are visible, some standing and others sitting, suggesting a collaborative workspace or a public area. The overall color palette is muted, with greys, blues, and the warm tones of the light coming through the glass.

Algorithms Problem Solving

Level 2

Check Number In Array

Definition

- ◆ This program fills an array with random numbers from `1` to `100`.
- ◆ It asks the user for a number to search for.
- ◆ It checks whether the number exists in the array.
- ◆ `IsNumbInArray()` returns `true` if the number exists and `false` if it does not.
- ◆ The program then prints whether the number was found.

Explanation

- ◆ ``FindNumberPositionInArray()`` searches the array and returns the index of the first occurrence.
- ◆ If the number is not found, it returns ``-1``.
- ◆ ``IsNumbInArray()`` uses the result of ``FindNumberPositionInArray()`` to determine whether the number exists.
- ◆ The intended condition should be:
 `return FindNumberPositionInArray(arr, arrLenght, numb) != -1;`
- ◆ This means:
 `index != -1` → number exists → true
 `index == -1` → number does not exist → false
- ◆ The current code has a mistake:
 `return FindNumberPositionInArray(arr, arrLenght, numb) != 1;`
- ◆ It compares the returned index with ``1`` instead of ``-1``.
- ◆ Also, ``IsNumbInArray()`` is not actually called in ``main()`` in this version of the program.

Steps in main()

1. Ask the user for the number of elements.
2. Fill the array with random numbers from `1` to `100`.
3. Print the array elements.
4. Ask the user for a number to search for.
5. Call `FindNumberPositionInArray()`.
6. Search through the array.
7. Return the index if the number is found.
8. Return `-1` if the number is not found.
9. Use the returned value to determine whether the number exists.
10. Print whether the number was found.

💡 Example

◆ `Array Elements` :

19 70 18 86 70 9 16 23

◆ Number to search:

19

◆ `FindNumberPositionInArray()` finds `19` at index `0`.

◆ Therefore:

$0 \neq -1$

◆ The result is `true`, meaning the number exists.

➡ Result

The number 19 is found :-)

✚ Note

◆ The main idea of this exercise is to reuse a function that returns an index to create another function that returns a `bool`.

◆ `FindNumberPositionInArray()` answers:

"Where is the number?"

◆ `IsNumbInArray()` answers:

"Does the number exist?"

◆ The corrected function is:

```
bool IsNumbInArray(int arr[], int arrLenght, int numb)
{
    return FindNumberPositionInArray(arr, arrLenght, numb) != -1;
}
```

◆ This is useful because the caller does not need to know the exact position when it only needs to know whether the number exists.

⊘ Warning

- ◆ There is a bug in the current `IsNumbInArray()` function: `!= 1` should be `!= -1`.
- ◆ `IsNumbInArray()` is currently unused in `main()`.
- ◆ `ReadNumber()` does not prevent the user from entering more than `100` elements.
- ◆ `cmath` and `corecrt_math_defines.h` are unnecessary.
- ◆ `srand()` and `time()` should ideally have their proper headers included explicitly.
- ◆ If the number appears multiple times, `FindNumberPositionInArray()` returns the index of the first occurrence.



Comparison

▣ `FindNumberPositionInArray()`

- ✓ Returns an `int`.
- ✓ Returns the position/index of the number.
- ✓ Returns `-1` when the number is not found.

▤ `IsNumbInArray()`

- ✓ Returns a `bool`.
- ✓ Returns `true` when the number exists.
- ✓ Returns `false` when the number does not exist.
- ✓ Does not need to return the actual position.

✦✦ Summary

- ✓ `FindNumberPositionInArray()` finds the position of a number.
- ✓ `-1` means the number was not found.
- ✓ `IsNumbInArray()` should return `true` when the position is not `-1`.
- ✓ Correct condition: `!= -1`.
- ✓ `IsNumbInArray()` is useful when we only need to know whether the number exists.

Thank you 😊